

Abstract

 1 The Internet of Things has instrumented billions of homes, buildings, farms, and factories with smart devices, yet today’s platforms fail to meet smart-space users’ needs. Control logic is concentrated in vendor clouds and hubs that add latency, cost, and single points of failure; fire-and-forget remote procedure calls (RPCs) leave developers and occupants without feedback on whether actions take effect; and users navigate a control interface spectrum with no structured support. This thesis argues that smart-space control should be designed around the needs of the people who occupy the space, across three planes of the smart-space stack—the *device action*, *smart-space automation*, and *human interaction* planes.

We propose new abstractions and insights at each of these three planes, spanning distributed systems and human-computer interaction. On the *device action plane*, RASC (Request-Ack-Start-Complete) replaces fire-and-forget RPC with an abstraction that exposes the full action lifecycle atop existing platforms, so automations and occupants can observe and program what their devices do; proofs and experiments confirm fast, low-overhead lifecycle detection and large scheduling gains over state-of-the-art baselines. On the *smart-space automation plane*, CoMesh removes the hub bottlenecks, cloud dependency, and outages of today’s centralized control: it is the first fully decentralized, hub-less control plane for large commercial edge IoT meshes, using group-based coordination with zero-message exchange and locality-sensitive hashing to keep control local, scalable, and fault-tolerant. On the *human interaction plane*, we perform a study of real smart-home owners via surveys and interviews, and derive insights showing that control is a fluid, situational process shaped by reliability and cognitive load rather than ecosystem boundaries, motivating how future smart-space operating systems should actually be built. Though independent, the three contributions share one vision: each makes a plane of smart-space control resilient, observable, and built around real user needs.

Summary of Comments on Human-Centered Control of Smart Spaces

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Author: Emily Wuchner

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